



Inner Product Space

Department of Computer Engineering
Sharif University of Technology

Hamid R. Rabiee rabiee@sharif.edu

Maryam Ramezani maryam.ramezani@sharif.edu



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01

Linear Form



What are Linear Functions?

- $f: R^n \rightarrow R$ means that f is a function that maps real n -vectors to real numbers
- $f(x)$ is the value of function f at x (x is referred to as the argument of the function).
- $f(x) = f(x_1, x_2, \dots, x_n)$: where $x_1, x_2, \dots, x_n$ are arguments

Definition

A function $f: R^n \rightarrow R$ is linear if it satisfies the following two properties:

- **Additivity:** For any n -vector x and y , $f(x + y) = f(x) + f(y)$
- **Homogeneity:** For any n -vector x and any scalar $\alpha \in R$: $f(\alpha x) = \alpha f(x)$

Superposition property:

Definition

Superposition property:

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

Note

❑ A function that satisfies the superposition property is called **linear**

What are Linear Functions?

- If a function f is linear, superposition extends to linear combinations of any number of vectors:

$$f(\alpha_1 x_1 + \cdots + \alpha_k x_k) = \alpha_1 f(x_1) + \cdots + \alpha_k f(x_k)$$

Inner product is Linear Function?

Theorem 1

A function defined as the inner product of its argument with some fixed vector is linear.

Proof: $f(x) = a^T x = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n$

$$\begin{aligned} 1) f(x+y) &= a^T(x+y) = a_1(x_1+y_1) + \cdots + a_n(x_n+y_n) \\ &= a_1 x_1 + \cdots + a_n x_n + a_1 y_1 + \cdots + a_n y_n = a^T x + a^T y = f(x) + f(y) \end{aligned}$$

$$\begin{aligned} 2) \alpha f(x) &= \alpha(a^T x) = \alpha(a_1 x_1 + \cdots + a_n x_n) = a_1 \alpha x_1 + \cdots + a_n \alpha x_n \\ &= a^T (\alpha x) = f(\alpha x) \end{aligned}$$

What are Linear Functions?

Theorem 2

If a function is linear, then it can be expressed as the inner product of its argument with some fixed vector.

Proof:

$$f(x) = f(x_1 e_1 + \cdots + x_n e_n) = x_1 f(e_1) + \cdots + x_n f(e_n)$$

$$= \begin{pmatrix} f(e_1) \\ \vdots \\ f(e_n) \end{pmatrix}^\top \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} f(e_1) \\ \vdots \\ f(e_n) \end{pmatrix}^\top x$$

What are Linear Functions?

Theorem 3

The representation of a linear function f in a specific basis as:
 $f(x) = a^T x$ is **unique**, which means that there is only one vector a for which $f(x) = a^T x$ holds for all x .

Proof:

$$\begin{aligned} f(x) &= a^T x, & f(x) &= b^T x \\ f(e_1) &= a^T e_1 = a_1, & f(x) &= b^T e_1 = b_1 \Rightarrow a_1 = b_1 \\ f(e_2) &= a^T e_2 = a_2, & f(x) &= b^T e_2 = b_2 \Rightarrow a_2 = b_2 \\ & & \dots & \\ f(e_n) &= a^T e_n = a_n, & f(x) &= b^T e_n = b_n \Rightarrow a_n = b_n \\ & & \Rightarrow a &= b \end{aligned}$$

Linear Form Examples

Example

- Is average a linear function?
- Is maximum a linear function?

$$\text{avg}(x) = \frac{x_1 + \cdots + x_n}{n} = \frac{1}{n}x_1 + \cdots + \frac{1}{n}x_n = \begin{pmatrix} \frac{1}{n} \\ \vdots \\ \frac{1}{n} \end{pmatrix}^T x$$

$$x = \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} \max(x) = 5, \quad y = \begin{pmatrix} 0 \\ 7 \\ 8 \end{pmatrix} \max(y) = 8$$

$$\max(x + y) = 11 \neq 5 + 8 = 13$$

02

Bilinear Form



Bilinear Form over a real vector space

Definition

Suppose V and W are vector spaces over the same field $\mathbb{F}$. Then a function $f: V \times W \rightarrow \mathbb{F}$ is called a **bilinear form** if it satisfies the following properties:

a) It is linear in its first argument:

- i. $f(\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}) = f(\mathbf{v}_1, \mathbf{w}) + f(\mathbf{v}_2, \mathbf{w})$ and
- ii. $f(c\mathbf{v}_1, \mathbf{w}) = cf(\mathbf{v}_1, \mathbf{w})$ for all $c \in \mathbb{F}, \mathbf{v}_1, \mathbf{v}_2 \in V$, and $\mathbf{w} \in W$.

b) It is linear in its second argument:

- i. $f(\mathbf{v}, \mathbf{w}_1 + \mathbf{w}_2) = f(\mathbf{v}, \mathbf{w}_1) + f(\mathbf{v}, \mathbf{w}_2)$ and
- ii. $f(\mathbf{v}, c\mathbf{w}_1) = cf(\mathbf{v}, \mathbf{w}_1)$ for all $c \in \mathbb{F}, \mathbf{v} \in V$, and $\mathbf{w}_1, \mathbf{w}_2 \in W$.

Bilinear Form

Note

Let V be a vector space over a field $\mathbb{F}$. Then the **dual** of V , denoted by V^* , is the vector space consisting of all linear forms on V .

Example

Let V be a vector space over a field $\mathbb{F}$. Show that the function $g: V^* \times V \rightarrow \mathbb{F}$ defined by

$$g(f, v) = f(v) \text{ for all } f \in V^*, v \in V$$

is a bilinear form.

Example Solution!

- $g(f_1 + f_2, v) = (f_1 + f_2)v = f_1(v) + f_2(v) = g(f_1, v) + g(f_2, v)$
- $g(cf, v) = (cf)(v) = cf(v) = cg(f, v)$
- $g(f, v_1 + v_2) = f(v_1) + f(v_2) = g(f, v_1) + g(f, v_2)$
- $g(f, cv) = f(cv) = cf(v) = cg(f, v)$

Symmetric Bilinear Form

Definition

A **bilinear form** function $f: V \times V \rightarrow \mathbb{F}$ over a real vector space V is called **symmetric** if for all $v, w \in V$:

$$f(v, w) = f(w, v)$$

Bilinear Form arises from a matrix

Theorem 4

Every **bilinear form** function $f: V \times V \rightarrow \mathbb{F}$ over a real vector space V arises from a matrix for all $v, w \in V$:

$$f(v, w) = v^T A w$$

Proof?

$$\begin{aligned} f(v, w) &= f\left(\sum_i v_i e_i, \sum_j w_j e_j\right) = \sum_i \sum_j v_i w_j f(e_i, e_j) \\ &= \sum_i \sum_j v_i a_{ij} w_j = v^T A w \end{aligned}$$

Associated Matrices

Definition

If V is a finite-dimensional vector space, $B = \{b_1, \dots, b_n\}$ is a basis of V , and $f: V \times V \rightarrow \mathbb{F}$ be a **bilinear form** function the **associated matrix** A of f with respect to B is the matrix $[f]_B \in \mathbb{F}^{n \times n}$ whose (i, j) -entry is the value $f(b_i, b_j)$.

$$f(v, w) = v^T A w = v^T [f]_B w$$

$$[f]_B = \begin{pmatrix} f(b_1, b_1) & \dots & f(b_1, b_n) \\ \vdots & & \vdots \\ f(b_n, b_1) & \dots & f(b_n, b_n) \end{pmatrix}$$

Associated Matrices

Note

The associated matrix changes if we use a different basis.

Example

For the bilinear form $f\left(\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix}\right) = 2ac + 4ad - bc$ on $\mathbb{F}^2$, find $[f]_B$ for basis $B = \left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \end{bmatrix}\right\}$ and $[f]_P$ for basis $P = \left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$

$$[f]_B = \begin{bmatrix} 14 & 29 \\ -16 & 20 \end{bmatrix}$$

$$[f]_P = \begin{bmatrix} 2 & 4 \\ -1 & 0 \end{bmatrix}$$

03

Bilinear Form Over Complex Vector Space



Bilinear Form over a complex vector space

Definition

Suppose V and W are vector spaces over the same field $\mathbb{C}$. Then a function $f: V \times W \rightarrow \mathbb{C}$ is called a **bilinear form** if it satisfies the following properties:

a) It is **linear in its first argument**:

- i. $f(\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}) = f(\mathbf{v}_1, \mathbf{w}) + f(\mathbf{v}_2, \mathbf{w})$ and
- ii. $f(\lambda \mathbf{v}_1, \mathbf{w}) = \lambda f(\mathbf{v}_1, \mathbf{w})$ for all $\lambda \in \mathbb{C}, \mathbf{v}_1, \mathbf{v}_2 \in V$, and $\mathbf{w} \in W$.

b) It is **conjugate linear in its second argument**:

- i. $f(\mathbf{v}, \mathbf{w}_1 + \mathbf{w}_2) = f(\mathbf{v}, \mathbf{w}_1) + f(\mathbf{v}, \mathbf{w}_2)$ and
- ii. $f(\mathbf{v}, \lambda \mathbf{w}_1) = \bar{\lambda} f(\mathbf{v}, \mathbf{w}_1)$ for all $\lambda \in \mathbb{C}, \mathbf{v} \in V$, and $\mathbf{w}_1, \mathbf{w}_2 \in W$.

Bilinear Form over a complex vector space



Bilinear forms on $\mathbb{R}^n$	Bilinear forms on $\mathbb{C}^n$
<u>Linear</u> in the first variable	<u>Linear</u> in the first variable
<u>Linear</u> in the second variable	<u>Conjugate linear</u> in the second variable



04

Inner Product



Inner product over real vector space

Definition

An inner product is a **positive-definite** **symmetric** **bilinear** form.

An inner product on V is a function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ such that $v, w \in V, c \in \mathbb{R}$:

1. $\langle v, v \rangle = 0$ if and only if $v = 0$.
2. $\langle w, v \rangle = \langle v, w \rangle$.
3. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$.
4. $\langle cw, u \rangle = c\langle w, u \rangle$ for all $u, w \in V$ and $c \in \mathbb{R}$.
5. $\langle v, v \rangle \geq 0$ for all $v \in V$.

Inner Product

Why for bilinear form I wrote just two properties instead of four properties?

- Using properties (2) and (4) and again (2)

$$\langle w, cu \rangle = \langle cu, w \rangle = c\langle u, w \rangle = c\langle w, u \rangle$$

- Using properties (2), (3) and again (2)

$$\langle w, u + v \rangle = \langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle = \langle w, u \rangle + \langle w, v \rangle$$

1. $\langle v, v \rangle = 0$ if and only if $v = 0$.
2. $\langle w, v \rangle = \langle v, w \rangle$.
3. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$.
4. $\langle cw, u \rangle = c\langle w, u \rangle$ for all $u, w \in V$ and $c \in \mathbb{R}$.
5. $\langle v, v \rangle \geq 0$ for all $v \in V$.

Inner Products

Note

□ For $v \in V$, $\langle 0, v \rangle = 0$, $\langle v, 0 \rangle = 0$.

$$\langle 0, v \rangle = \langle 0u, v \rangle = 0\langle u, v \rangle = 0$$

General Inner product

Definition

Suppose that $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ and that V is a vector space over $\mathbb{F}$. Then an **inner product** on V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ such that the following three properties hold for all $c \in \mathbb{F}$ and all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in V$:

- a) $\langle \mathbf{v}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{v} \rangle}$ (conjugate symmetry)
- b) $\langle \mathbf{v} + c\mathbf{x}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle + c\langle \mathbf{x}, \mathbf{w} \rangle$ (linearity)
- c) $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$. (pos. definiteness)

Inner Products for vectors

Note

- The standard inner product between vectors is: $(x, y \in \mathbb{R}^n)$

$$\langle x, y \rangle = x^T y = \sum x_i y_i$$

- The function $\langle \cdot, \cdot \rangle : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ defined by

$$\langle v, w \rangle = v^* w = \sum_{i=1}^n \bar{v}_i w_i$$

for all $v, w \in \mathbb{C}^n$ is an inner product on $\mathbb{C}^n$.

Inner Products for matrices

Note

The standard inner product between two matrices is: $(X, Y \in \mathbb{R}^{m \times n})$

$$\langle X, Y \rangle = \text{trace}(X^T Y) = \sum_i \sum_j X_{ij} Y_{ij}$$

Example

Find the inner products of following matrices:

$$U = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\langle U, V \rangle = \text{trace}(U^T V) = \text{trace} \left(\begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix} \right) = 0$$

Inner Product for functions

Note

Let $a < b$ be real numbers and let $C[a, b]$ be the vector space of continuous functions on the real interval $[a, b]$. The function $\langle \cdot, \cdot \rangle : C[a, b] \times C[a, b] \rightarrow \mathbb{R}$ defined by

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx \quad \text{for all } f, g \in C[a, b]$$

is an inner product on $C[a, b]$.

Inner Product for polynomials

Note

□ For $p(x)$ and $q(x)$ with at most degree n :

$$\langle p(x), q(x) \rangle = p(0)q(0) + p(1)q(1) + \cdots + p(n)q(n)$$

□ For $p(x)$ and $q(x)$: $\langle p(x), q(x) \rangle = p(0)q(0) + \int_{-1}^1 p'q'$

□ For $p(x)$ and $q(x)$: $\langle p(x), q(x) \rangle = \int_0^\infty p(x)q(x)e^{-x}dx$

05

Inner Product Space

Inner product space

Definition

An **inner product space** is a finite-dimensional real or complex vector space V along with an inner product on V .

Euclidean Space Unitary Space

06

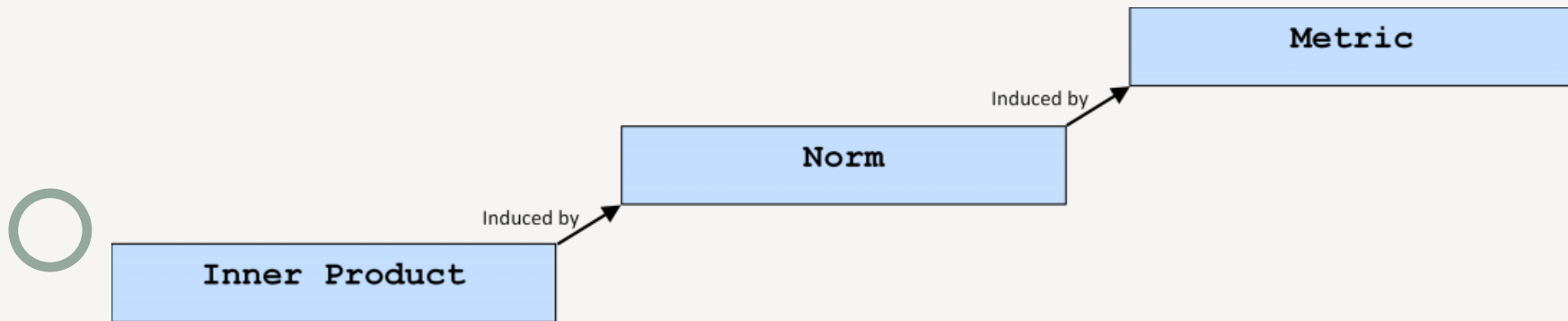
Norm Space



The reason to use norms

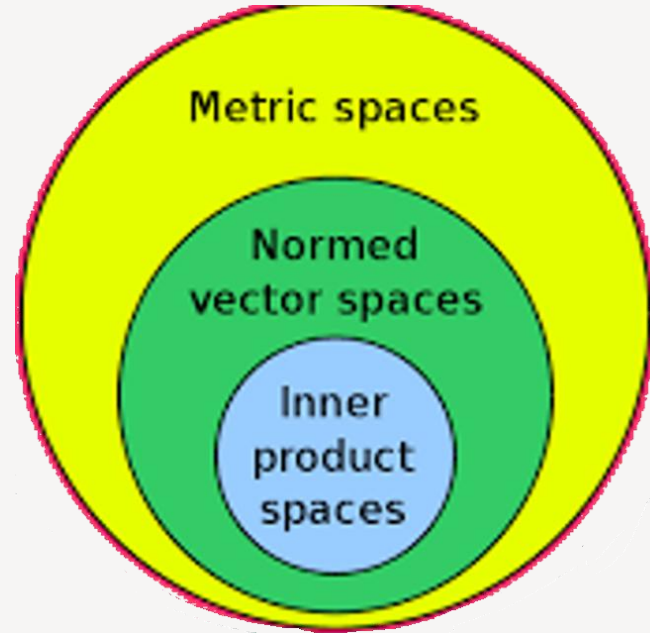
- ❑ Machine learning uses vectors, matrices, and tensors as the basic units of representation
- ❑ Two reasons to use norms:
 - To estimate how **big** a vector/matrix/tensor is
 - How big is the difference between two tensors is
 - To estimate how **close** one tensor is to another
 - How close is one image to another

Inner Products, Norms and Metrics



Inner Products, Norms and Metrics

- Given an inner product $\langle A, B \rangle$, one can obtain a norm doing
$$\|A\|^2 = \langle A, A \rangle$$
- And given a norm $\|A\|$, one can obtain a metric using the difference vector $\|A - B\|$



Inner Products, Norms and Metrics



Vector space	Generalization
metric	metric space
norm	normed
scalar product	inner product space



Euclidean Norm

Definition

Functions closely related to inner products are so-called **norms**. Norms are specific functions that can be interpreted as a distance function between a vector and the origin.

Definition

For $v \in V$, we define the euclidean norm of v , denoted $||v||$, by:

$$||v|| = \sqrt{\langle v, v \rangle}$$

Euclidean Norm

Note

- Euclidean Norm (2-norm, l_2 norm, length)

$$||x|| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{x^T x}$$

- A vector whose length is 1 is called a **unit vector**
- **Normalizing**: divide a non-zero vector by its length which is a unit vector in the same direction of original vector
- It is a nonnegative scalar
- In $\mathbb{R}^2$ follows from the Pythagorean Theorem.
- What about $\mathbb{R}^3$?
- What is the shape of $||x||_2 = 1$?

Euclidean Norm

Example

Norm of $P_n(x)$ in the term of inner product $\langle p_n(x), q_n(x) \rangle = \int_0^1 p_n(x)q_n(x)dx$:

$$||P_n(x)|| = \sqrt{\int_0^1 P_n^2(x)dx}$$

Vector Norm Properties

Important Properties

1. Absolute Homogeneity / Linearity:

$$||\alpha x|| = |\alpha| ||x||$$

2. Subadditivity / Triangle Inequality:

$$||x + y|| \leq ||x|| + ||y||$$

3. Positive definiteness / Point separating:

$$\text{if } ||x|| = 0 \text{ then } x = 0$$

(from 1 & 3): For every x , $||x|| = 0$ iff x

$= 0$

4. Non-negativity:

$$||x|| \geq 0$$

Norm of sum

Theorem 4

If x and y are vectors:

$$||x + y|| = \sqrt{||x||^2 + 2 x^T y + ||y||^2}$$

Proof:

$$\begin{aligned} ||x + y||^2 &= (x + y)^T (x + y) \\ &= x^T x + x^T y + y^T x + y^T y \\ &= ||x||^2 + 2 x^T y + ||y||^2 \end{aligned}$$

Inner product and norm

Theorem 5

Take any inner product $\langle \cdot, \cdot \rangle$ and define $f(x) = \sqrt{\langle x, x \rangle}$. Then f is a norm.

Note

Every inner product gives rise to a norm, but not every norm comes from an inner product. (Think about norm 2 and norm max)

Proof

- **Positive definiteness**

Inner product properties give:

$$\langle x, x \rangle \geq 0 \text{ for all } x, \text{ so } f(x) = \sqrt{\langle x, x \rangle} \geq 0.$$

$$\langle x, x \rangle = 0 \text{ iff } x = 0.$$

$$\text{Hence } f(x) = 0 \iff x = 0.$$

- **Homogeneity (absolute scalability)**

For any scalar α ,

$$f(\alpha x) = \sqrt{\langle \alpha x, \alpha x \rangle} = \sqrt{\alpha \bar{\alpha} \langle x, x \rangle} = \sqrt{|\alpha|^2 \langle x, x \rangle} = |\alpha| \sqrt{\langle x, x \rangle} = |\alpha| f(x).$$

(For real spaces, $\alpha \bar{\alpha} = \alpha^2$.)

Proof (cont')

- **Triangle inequality**

Use Cauchy-Schwarz:

$$|\langle x, y \rangle| \leq f(x) f(y).$$

Now expand:

$$f(x + y)^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, y \rangle + \langle x, y \rangle + \langle y, x \rangle$$

In real spaces, $\langle y, x \rangle = \langle x, y \rangle$, so

$$\begin{aligned} f(x + y)^2 &= f(x)^2 + f(y)^2 + 2\langle x, y \rangle \leq f(x)^2 + f(y)^2 + 2|\langle x, y \rangle| \\ &\leq f(x)^2 + f(y)^2 + 2f(x)f(y) = (f(x) + f(y))^2 \end{aligned}$$

Taking square roots (both sides ≥ 0):

$$f(x + y) \leq f(x) + f(y).$$

- All three norm axioms hold, so f is a norm. $\square$

Metric Properties

Important Properties

Let V be a real vector space over $\mathbb{R}$. A function $V \times V \rightarrow \mathbb{R}$ is called **metric** or **distance function** on V , and (V, d) a metric space, if for all $u, v, w \in V$ the following holds true:

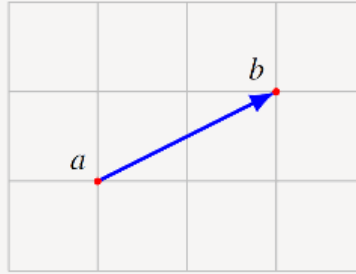
(i) $d(v, w) \geq 0$ and $d(v, w) = 0$ if and only if $v = w$;

(ii) $d(v, w) = d(w, v)$;

(iii) $d(v, w) \leq d(v, u) + d(u, w)$.

Euclidean Distance

- Distance between two n-vectors shows the vectors are “close” or “nearby” or “far”.



- Distance:

$$\text{dist}(a, b) = ||a - b||$$

Comparing Norm and Distance

Norm

(Normed Linear Space)

- 
1. $\|x - y\| \geq 0$
 2. $\|x - y\| = 0 \Rightarrow x = y$
 3. $\|\lambda(x - y)\| = |\lambda| \|x - y\|$

Distance Function

(Metric Space)

1. $\text{dist}(x, y) \geq 0$
2. $\text{dist}(x, y) = 0 \Rightarrow x = y$
3. $\text{dist}(x, y) = \text{dist}(y, x)$

Resources

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